ScratchVM:

understanding virtualization from scratch

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Introduction:
Notions of Virtualization

Definitions

*"**Virtualization** is the application of the layering principle through enforced modularity, whereby the exposed virtual resource is identical to the underlying physical resource being virtualized."**

*"**A virtual machine** is an abstraction of a complete compute environment through the combined virtualization of the processor, memory, and I/O components of a computer."**

*"**The hypervisor** is a specialized piece of system software that manages and runs virtual machines."**

*"**The virtual machine monitor (VMM)** refers to the portion of the hypervisor that focuses on the CPU and memory virtualization."**

Hardware and Software Support for Virtualization*. Pr. Edouard Bugnion, Pr. Jason Nieh & Dr. Dan Tsafrir. **Synthesis Lectures on Computer Architecture. 2017

Overview: 3 software layers

- VMM: Virtual Machine Monitor (e.g., Qemu, Firecracker)
  - Management (creation, destruction, monitoring, etc…)
  - Emulation (instructions, devices, etc…)
  - Runs in user mode on the host
- Hypervisor (e.g., Linux KVM)
  - Multiplexes and configures resources
  - Emulation (instructions, devices, etc…)
  - Runs with host highest privileges
- Virtual Machine
  - Isolated environment from host machine
  - Can (only) access resources provided by the host (VMM or Hypervisor)
  - Composed of **virtual CPU** (1 or more) and **physical memory**

Linux Kernel Virtual Machine (KVM)

KVM: Linux Kernel Virtual Machine Module

- Linux Kernel module (“/dev/kvm”)
  - Check if enabled with **kvm-ok** command
  - installation procedure
- VMM KVM service through ioctl system calls
  - Getting the file descriptor with **open** syscall
  - ioctl(kvmfd, KVM_SERVICE, &arguments);
  - KVM API documentation
- Necessary for virtual machine management
  - **physical memory**
  - **virtual CPU**
  - devices
  - etc…

Part I:
Create and run a x86 binary program

KVM: Physical Memory

- Associated to host virtual memory
  - Guest Physical Address → Host Virtual Address
  - Allocated with **mmap** syscall by the VMM on the host
- KVM memory region describes guest physical memory area
  - Starting (Guest Physical) Address
  - Memory Size
  - Corresponding Host Virtual Address
  - KVM service: **KVM_SET_USER_MEMORY_REGION**

KVM: Virtual CPU

- Privileged registers:
  - Control Registers (CR3, ...)
  - Model Specific Registers (MSR)
  - ...
- Unprivileged registers:
  - Regular Registers (r1-15)
  - Instruction Pointer Register (RIP)
  - Stack Pointer Register (RSP)
  - ...